⚡ PROBLEM STA… NTUM DAY 2026

[Problem Statement Title]

Company Logo

[Company Name]

[Company tagline or domain]

EVENT DATE
14 April 2026

ORGANIZED BY
IIT Indore & NQM

DIFFICULTY
[Level]

📋 Instructions for Speakers

Thank you for participating in **"What Quantum Industry Wants?"** — an initiative by **IIT Indore** and the **National Quantum Mission** to bridge the gap between quantum industry and academia by making students industry ready before the placement/joining.

Please fill in the sections marked with *[placeholder text]* in this template. The completed document will be shared on our **dedicated Discord channel** where interested 3rd-year undergraduate students can collaborate to solve your problem statement over the summer.

Outcome: Students who successfully solve the problem statement may receive an **internship or pre-placement offer (PPO)** from your organization.

⚠ **Disclaimer:** This event is intended solely to connect quantum industry leaders with interested students. Neither IIT Indore nor the host (Dr. Shashank Gupta) is responsible for the evaluation of submissions or the quality of solutions provided to the problem statements. All evaluation and hiring decisions rest entirely with the respective industry partner.

TEMPLATE STRUCTURE

1. **About Your Company** — Brief introduction to your organization and its quantum focus
2. **Motivation** — Why this problem matters and the real-world impact
3. **Problem Statement** — Detailed description of the challenge to be solved
4. **Requirements** — Technical specifications, constraints, and scope
5. **Deliverables** — What participants must submit
6. **Evaluation & Resources** — Judging criteria and helpful references

ABOUT US

Introduction & Company Overview

Provide a brief overview of your company, its mission, and its work in the quantum technology space.

[Write 2–3 paragraphs about your company here. Include:
- *Company founding year and headquarters*
- *Core technology focus and products/services*
- *Key achievements, patents, or milestones*
- *Team size and notable leadership*
- *Vision for the future of quantum technology]*

Core Focus Area
[e.g., Quantum Computing / Cryptography / Sensing / Communication]

Headquarters
[City, Country]

Founded
[Year]

Website
[Company URL]

MOTIVATION

Why This Problem Matters

Explain the real-world significance of this problem. Why should students care? What impact will solving it have on the industry or society?

[Describe the motivation behind this problem statement. Consider:
- *What industry need or gap does this address?*
- *What are the current limitations or challenges?*
- *How does this connect to the broader quantum ecosystem?*
- *What makes this problem timely and important?]*

Impact Statement: *[Describe in 1–2 sentences the real-world impact of solving this problem]*

PREREQUISITES

What knowledge or skills should students ideally have before attempting this problem?

[List recommended prerequisites, e.g.:
- *Quantum mechanics fundamentals*
- *Programming languages (Python, C/C++, etc.)*
- *Specific frameworks or libraries*
- *Linear algebra / cryptography background]*

PROBLEM STATEMENT

Detailed Problem Description

Provide a clear and detailed description of the challenge. Be specific about what the students need to build, implement, or solve.

[Write a comprehensive problem statement. Include:
- *Clear objective — what must be achieved?*
- *Technical context — what systems/technologies are involved?*
- *Constraints — what limitations must be respected?*
- *Scope — what is in scope vs. out of scope?*

Be specific enough that a motivated 3rd-year undergraduate can understand the problem and start working on it independently.]

KEY CHALLENGES

List the core technical challenges students will face.

01 *[Challenge 1 Title]*
[Brief description of this challenge]

02 *[Challenge 2 Title]*
[Brief description of this challenge]

03 *[Challenge 3 Title]*
[Brief description of this challenge]

REQUIREMENTS

Technical Specifications

Define the technical requirements, constraints, and specifications that students must follow.

Technical Stack & Tools

[Specify required/recommended:
- *Programming languages*
- *Frameworks & libraries*
- *Hardware platforms (if any)*
- *Cloud services or simulators]*

Constraints

[List technical constraints:
- *Performance requirements*
- *Memory/resource limits*
- *Compatibility requirements]*

Acceptance Criteria

[Define what constitutes a valid solution:
- *Minimum benchmarks*
- *Required functionality*
- *Quality standards]*

Note: Students will be working on these problem statements over the summer break (May – July 2026). Please calibrate the scope accordingly — the problem should be **challenging yet achievable** within 8–10 weeks of focused effort by a team of 2–4 undergraduates.

DELIVERABLES

What Students Must Submit

Define exactly what participants need to deliver upon solving the problem statement.

01 **Source Code & Repository**
[Specify form … ure requirements.]

02 **Technical Report**
[Specify len … ethodology, results, analysis.]

03 **Demo / Presentation**
[Specify if a vi … clude duration limits.]

04 **Additional Deliverables**
[Any additional submiss … atasets, etc.]

TIMELINE

Start Date
14 April 2026 (after the event)

Submission Deadline
[Suggested: July 2026]

Evaluation Period
[e.g., July – Aug 2026]

PPO Decisions
[Expected date for PPO offers]

EVALUATION CRITERIA

How Solutions Will Be Judged

Define the criteria and weightage for evaluating student submissions.

CRITERION	DESCRIPTION	WEIGHT
[Criterion 1]	[e.g., Correctness of implementation]	[%]
[Criterion 2]	[e.g., Performance & optimization]	[%]
[Criterion 3]	[e.g., Code quality & documentation]	[%]
[Criterion 4]	[e.g., Innovation & creativity]	[%]
[Criterion 5]	[e.g., Presentation & report quality]	[%]

RESOURCES & REFERENCES

Provide helpful resources, documentation, and references for the students.

- *[Resource 1: e.g., Documentation / Tutorial]* *[URL or reference]*
- *[Resource 2: e.g., GitHub Repository / SDK]* *[URL or reference]*
- *[Resource 3: e.g., Research Paper / Textbook]* *[URL or reference]*
- *[Resource 4: e.g., Tools / Simulator Access]* *[URL or reference]*

Contact for Queries: Students can reach out to *[Speaker Name]* at *[email]* for clarifications regarding this problem statement.

Mentor Availability: *[Specify if your company will provide mentorship/office hours during the summer]*

ORGANIZED BY
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